Faster Harvesting: Download Once, Read Locally

Nov 5, 2026 · @Garry Howard

In short

Downloading Wikidata once and reading it from our own disk should take the harvest from about 1 million facts an hour to a few hundred million, at almost no cost.

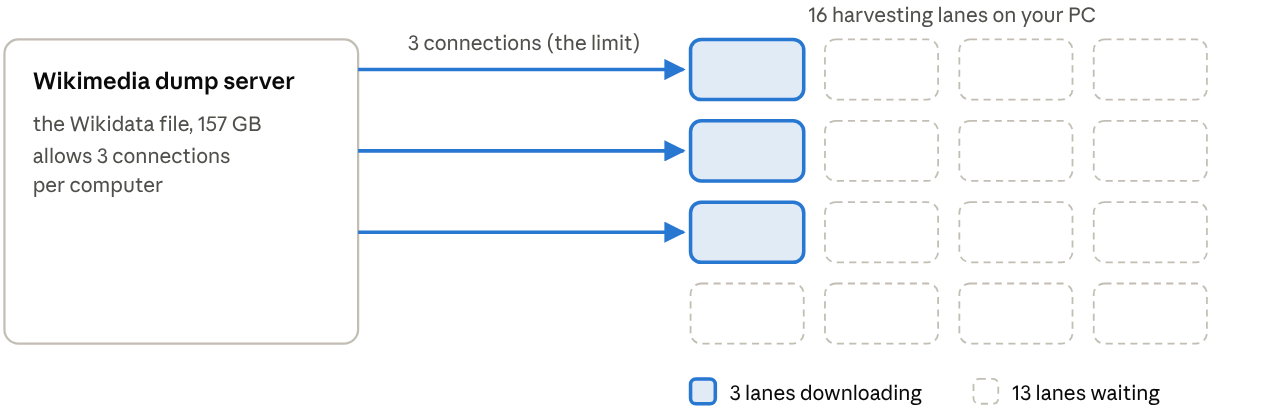
- **Today:** 16 harvesting lanes each fetch their own piece of Wikidata over the internet. Wikimedia allows only 3 connections per computer, so 13 lanes are always waiting in line. Measured result: about 1 million facts an hour, roughly 6 weeks to reach 1 billion.
- **New design:** one downloader fetches the whole Wikidata file (about 157 GB) using the 3 allowed connections, plus official mirror copies. Then all 16 lanes read that file from the local disk, where there are no limits. Estimated result on your 800 Mbps line: 1 billion facts in 3 to 5 hours.
- **Cost:** no new software, no subscriptions, no LLM tokens. Electricity for one afternoon of work, and about 660 GB of the 23.8 TB already free on D:.
- **What we will not do:** extra accounts, rotating IP addresses or rented containers to get around the limits. Wikimedia blocks clients that do this, and a block would stop the harvest completely.

All speeds and times here are estimates from measurements on this project's earlier runs, not yet tested on your computer.

Today: why the harvest crawls

The harvest is slow because the lanes share 3 internet connections, not because the computer is slow.

Today: 13 of 16 lanes wait in line for 3 connections



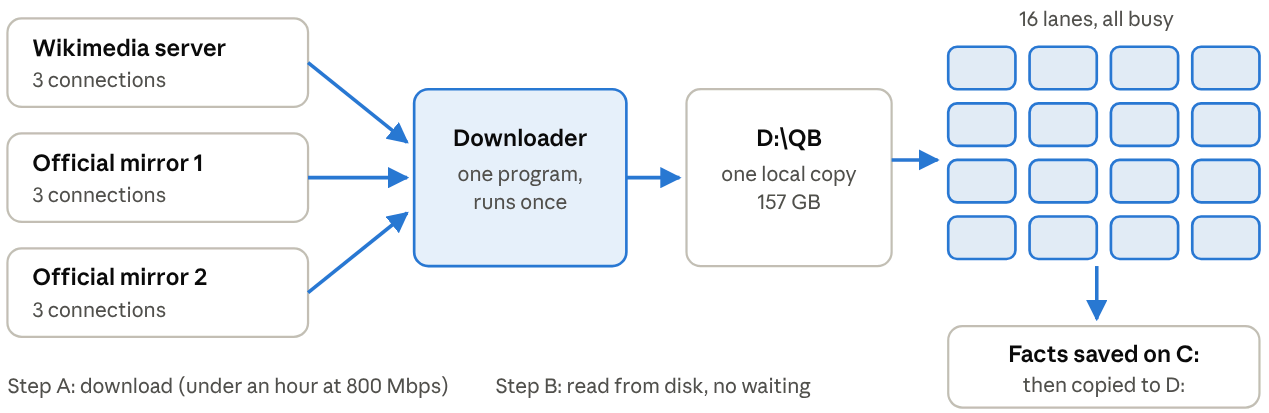
Result: about 1 million facts an hour (measured), about 6 weeks to 1 billion

today's harvest design · 16 lanes, 3 connections

Each lane downloads its own part of the file while it reads it. While 3 lanes download, the other 13 wait, and a lane that restarts must skip back through everything it already read. In an earlier analysis one lane spent 78% of its time skipping, and another waited 9 hours for a free connection.

The new design: download once, read locally

The fix is to separate the two jobs. First, one program fetches the whole file to the D: drive. Then every lane reads its share from that copy, at disk speed, with nobody waiting for the internet.



proposed design · 3 sources × 3 connections, then 16 local lanes (speeds are estimates)

Think of a library book. Today, 16 people each phone the library and have pages read to them, three calls at a time. In the new design, one person borrows the book once, photocopies it, and all 16 read their own copy at the same time.

The official mirrors are copies of the same file that universities and hosting companies offer on purpose, so downloading from them is welcome. Using two mirrors as well as the main server triples the download speed while staying within each site's 3-connection rule.

Workflow: from Start to a billion facts

You press Start once; QueryBook does the rest and shows each stage on the Harvesting tab.

1. **Check for a new edition.** QueryBook asks Wikimedia for the date and size of the latest Wikidata file (published weekly). If D:\QB already holds that edition, it skips straight to step 5.
2. **Pick the download sources.** The official Wikimedia server plus official mirrors (copies kept by universities and hosting companies). A mirror is used only if its file is byte-for-byte the same size and date.
3. **Download in pieces.** The file is split into pieces. Each source works on different pieces with at most 3 connections, so nothing breaks the per-computer limit. The tab shows percent done, speed in MB/s and time left.

4. **Check the file.** Every piece is checked as it lands. A broken or missing piece is fetched again from another source. Wikimedia's published checksum confirms the finished file.
5. **Read locally.** All 16 lanes read their own part of the file from D:\QB at full processor speed. Facts are written on C: first and copied to D:, as today.
6. **Remove duplicates and save.** Facts held twice are kept once, and the finished shards are saved to D:\QB.
7. **Keep up to date.** Each day QueryBook downloads Wikimedia's change file (about 0.7 GB) and applies only what changed, instead of starting again.

A download stopped part-way (power cut, Stop pressed) resumes from the last finished piece.

Every option compared

Downloading once from official mirrors is the fastest and cheapest option that stays within Wikimedia's rules.

Option	Facts per hour (estimate)	Time to 1 billion facts	Extra cost (estimate)	Allowed by Wikimedia	Risk
Download once from 2–3 official mirrors, read locally (recommended)	about 250–400 million	about 3–4 hours	$0 (electricity about $0.30)	Yes	Low: a mirror can be slow or out of date; it is then skipped
Download once from Wikimedia's own server, read locally	about 150–250 million	about 4–5 hours	$0	Yes	Low
Add an internal SSD for the working copies	the same, steadier	the same	about $80–150 for 2 TB, once	Not relevant	None
Run QLever (a fast Wikidata search engine) on this PC for the recurring-events queries	removes the query limit for that one dataset	not a bulk source	$0 software; about a day to set up	Yes	Low: needs disk space and memory
Wikimedia Enterprise (paid bulk access)	not needed once the dump is local	—	paid; price on request	Yes	None

Option	Facts per hour (estimate)	Time to 1 billion facts	Extra cost (estimate)	Allowed by Wikimedia	Risk
Today: 16 lanes streaming over 3 connections	about 1 million (measured)	about 1,000 hours (6 weeks)	about $75 electricity over 6 weeks	Yes	The harvest stays slow
Several accounts or IP addresses	depends on the number of addresses	unknown	residential proxies often cost $3–15 per GB moved	**No**	High: blocked, the harvest stops
Rented containers (for example Bunny.net Magic Containers)	depends on the number of containers	unknown	pay per CPU-hour and per GB moved; tens to hundreds of dollars per full harvest	**No**, when used to get round the limits	High: blocked, plus a monthly bill
Extra tablets or small PCs at home	no gain (they share one internet address)	about 1,000 hours	about $150–400 per device	Yes	Money spent for nothing

Cost figures are approximate and should be checked before any purchase. Electricity assumes a 150 W desktop at about $0.50 per kWh, a high tariff, because you said your rates are very high. Times assume your 800 Mbps connection.

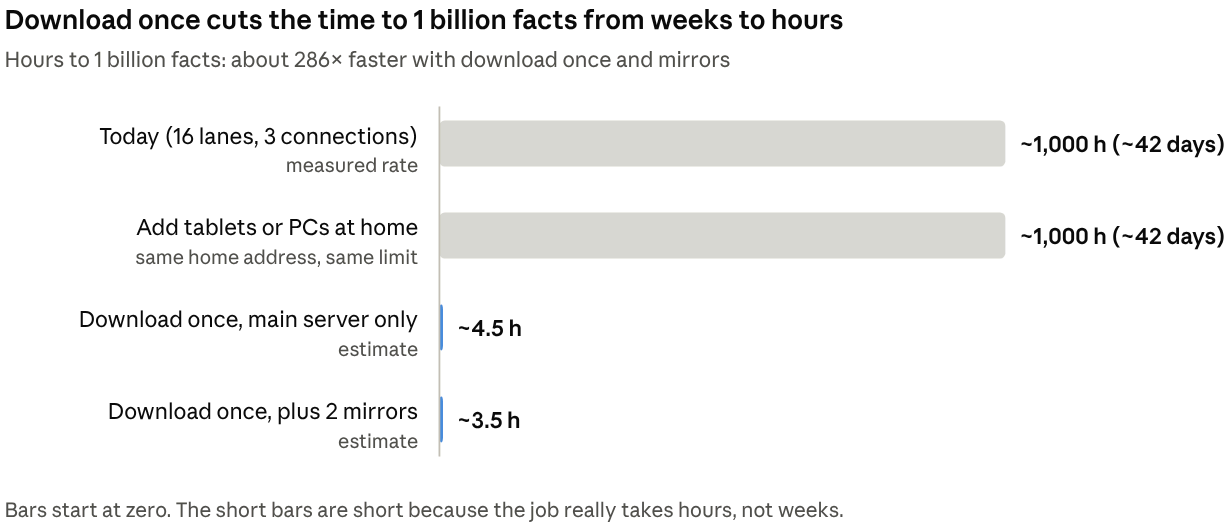
What we will not do, and why

Wikimedia's limits are per computer and per user, and its policy says clients that evade them may be blocked; once the file is on our disk the limits no longer matter anyway.

- **Several virtual clients with different accounts.** Splitting one harvest over several accounts is the evasion the policy describes. If noticed, every account and our address can be blocked, and the harvest stops completely.
- **Different IP addresses for each client.** Same problem, plus a running cost: residential proxy services charge per GB, and a full Wikidata harvest moves about 157 GB.
- **Rented containers (Bunny.net Magic Containers or similar).** Spreading the downloads over many rented addresses is the same evasion, at a higher price: you pay for each container's processor time and for every GB moved. They also gain nothing here: once the file is local, reading it needs no network at all.

- **More tablets or small PCs at home.** All devices behind one home router share one internet address, so they share the same 3 connections. Tablets also read the file far more slowly than a desktop processor.
- **LLM tokens.** Bulk harvesting from files needs no LLM at all. The LLM-assisted gatherer stays for small gaps only; its cost shows on the Control Panel's LLM usage page.

Cost estimates in detail



today's rate measured (about 1 million facts an hour); other times are estimates

All prices below are rough, in US dollars. They assume a home PC drawing about 150 watts, electricity at about $0.50 per kWh (a high tariff), and your 800 Mbps connection.

With high power prices, the slow way is the expensive way: today's harvest keeps the PC running for 6 weeks, while the new design needs it for one afternoon.

Item	One-off cost	Running cost	What it buys
Today's harvest (about 6 weeks of running)	$0	about $75 of electricity	1 billion facts in about 6 weeks
Download once (main server, or plus mirrors)	$0	about $0.30 of electricity	1 billion facts in 3 to 5 hours, unattended

Item	One-off cost	Running cost	What it buys
Optional: a fast internal SSD for the working copy	$80 to $150	$0	reading runs 2 to 5 times faster than on the USB drive
Bigger internet plan (not needed)	$0	$0	nothing: your 800 Mbps line already downloads the file in under an hour
Rented proxy addresses (not recommended)	$0	$3 to $15 per GB, so about $470 to $2,400 for one 157 GB copy	breaks the site rules and risks a block
Extra tablets or PCs (not recommended)	$150 to $400 each	small	nothing: they share the same home address and the same limit
Wikimedia Enterprise (paid feed)	$0	free trial tier; paid plans priced on request	a supported fast feed, if the free route is ever not enough

The cheapest option is also the fastest. Download once saves about $75 of electricity per billion facts compared with today, and the only purchase worth considering is the SSD.

Risks and what to watch

With your 800 Mbps line, the download takes under an hour, so reading the file now sets the pace. On a slower line, only the download time would grow.

Risk	What you would see	What QueryBook does
Slow internet (under 10 MB/s)	the download stage takes 5 hours or more	carries on; the time left is shown; nothing else changes
A mirror is slow, stale or offline	that source drops out of the download list	its pieces go to the other sources
The download stops (power cut, Stop pressed)	the stage shows "paused"	resumes from the last finished piece

Risk	What you would see	What QueryBook does
D: unplugged during the download	the stage waits	resumes when D: is back
Not enough room	an alert before the download starts	it needs about 660 GB free on D: (157 GB file, about 500 GB of facts); D: has 23.8 TB
The processor is the limit	facts per hour stop rising as lanes are added	normal: the network is no longer the limit

Words used here

- **Dump:** one big file holding everything Wikidata knows, published weekly.
- **Mirror:** an exact copy of that file kept by a university or hosting company, free to download.
- **Connection:** one download line to a server; Wikimedia allows 3 per computer.
- **Lane:** one QueryBook worker reading its own part of the file (16 run side by side).
- **Change file:** Wikimedia's daily list of what changed, about 0.7 GB.
- **QLever:** a fast Wikidata search engine that can run on this PC.